Safety-Certified Constrained Control of Maritime Autonomous Surface Ships for Automatic Berthing

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***Abstract*—The berthing of maritime autonomous surface ships (MASSs) is a challenging operation even for an experienced captain due to the required complicated maneuvering at low speeds in a constrained water space. This article addresses safety-aware automatic berthing of MASSs moving in a constrained water region with stationary and moving obstacles. A safety-critical constrained anti-disturbance control method is proposed for the automatic berthing of MASSs subject to velocity constraints, input constraints, and collision-avoidance constraints, as well as ocean disturbances. At the kinematic level, desired linear velocities and yaw rate are prescribed by incorporating a line-of-sight guidance scheme to achieve position-heading stabilization. At the kinetic level, an anti-disturbance kinetic control law is designed based on an extended state observer. Through converting the input constraints to velocity constraints on guidance signals, the safety-guidance signals are obtained by solving a quadratic programming problem subject to the state, input, and collision-avoidance constraints. It is proven that an MASS controlled by using the proposed method is input-to-state safe in the presence of ocean disturbances. Simulation results are elaborated to substantiate the efficacy of the proposed safety-certified control law for the automatic berthing of an MASS subject to physical constraints and environmental disturbances.**

***Index Terms*—Safety-certified control, automatic berthing, maritime autonomous surface ships, stationary and moving obstacles, input-to-state safety.**

I. Introduction

MARITIME autonomous surface ships (MASSs) are anticipated to be the major means in the future of maritime transport due to their improved efficiency, increased safety, and reduced seafarer cost [1], [2], [3], [4], [5], [6]. According to International Maritime Organization (IMO), MASSs are ships that can operate, to some degree, independently of human intervention. Ship berthing is traditionally conducted by experienced captains [7]. But for an MASS, automatic berthing is highly desirable or necessary when moving from open waters toward a docking area along the quay in a harbor [7], [8], [9], [10], [11]. From a control perspective, automatic berthing is a challenging task, which includes performing maneuvers to control a ship to a desired position and orientation, adhering to spatial constraints to avoid any collision in the presence of environmental disturbances [1], [2], [4], [7], [12], [13], [14], [15]. Existing automatic berthing methods range from reinforcement learning [7], artificial neural network [8], [16], [17], [18], fuzzy logic systems [19], model predictive control [9], [11], [20], [21], to adaptive dynamic programming [10], [22], [23], [24]. In [18], a control method is proposed for automatic berthing based on a neural network and the berthing performance is investigated under various initial conditions as well as wind disturbances. It is indicated that the control law may not be effective in keeping track of the trajectory due to wind disturbances [18]. In [12], an optimal control scheme with the capability of collision avoidance is proposed for automatic berthing, where the computational burden is demanding due to the use of a model predictive control method. In [13], a path-following control method is developed for automatic berthing and experimental results in both calm and strong wind conditions are provided to show its effectiveness, whereas safety constraints are not addressed. In [10], a heuristic dynamic programming method is proposed for automatic berthing, where input and safety constraints are not considered. In [24], a data-driven adaptive control method is proposed for automatic berthing and the experimental results are shown to demonstrate the feasibility and practicality. Despite the promising results in the recent literature, how to develop efficient control laws to achieve safety-aware automatic berthing in a constrained water region with stationary and moving obstacles is still open.

Motivated by the aforementioned results, in this study, we address the safety-aware automatic berthing of MASSs subject to velocity constraints, input constraints, and collision-avoidance constraints, in addition to model uncertainties and ocean disturbances. A safety-critical constrained anti-disturbance control method is proposed to achieve automatic berthing for the MASSs. At the kinematic level, desired linear velocities and yaw rate are obtained to achieve the position-heading stabilization based on line-of-sight (LOS) guidance. At the kinetic level, an anti-disturbance kinetic control law is developed based on an

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extended state observer for estimating total disturbance, such as model uncertainties and ocean disturbances. By taking account of the velocity constraints, input constraints, collision-avoidance constraints, a quadratic programming problem is formulated to obtain the safe guidance signals within these safety constraints. The advantages of the proposed method over the existing ones are discussed as follows.

- An optimization-based constrained anti-disturbance control method is developed to achieve optimal berthing performance without the need for training, in contrast to the learning-based automatic berthing control laws [7], [8], [9], [10], where the learning entails a large number of training data and a lengthy training process. As a result, the proposed method is less resource-consuming than the learning-based methods.
- The proposed method is able to address velocity constraints, input constraints, shoreline constraints, and obstacle constraints, leading to guaranteed safety in automatic berthing, in addition to closed-loop stability, without the need for path planning in advance as in [25], [26].
- The computational burden of the proposed method is drastically lower than those of the model predictive control scheme in [27], [28], [29] where iterative prediction and optimization are required.
- The proposed command optimization method directly optimizes the guidance signals subject to the constraints such that the control signals satisfy all constraints, thus ensuring safety from the control loop. In contrast, as the artificial potential field methods [30], [31], [32] address collision avoidance in the motion planning loop without considering state and input constraints, the prescribed signals from the motion planning loop may violate state and input constraints. As a result, the planned motion may not be implementable in the control loop.

The rest of this article is organized as follows. Section II introduces the ship kinetics and formulates the problem. Section III elaborates on the design and analysis of the safety-certified guidance and control law. Section IV provides simulation results for control performance illustrations. Finally, Section V concludes this article.

II. Preliminaries and Problem Formulation

Let $\Re^N$ denote as an N-dimensional Euclidean space, $|\cdot|$ as the absolute value of a real number, $\|\cdot\|$ as the 2-norm of a matrix or a vector, I_n as an $n\times n$ identity matrix.

A. Control Barrier Function

Consider a class of nonlinear systems in a control-affine form [33]:

$$\dot{x} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u}, \tag{1}$$

where $x \in \mathcal{D} \subset \Re^n$ is the state vector and $u \in \mathcal{U} \subset \Re^m$ is the admissible input vector. The functions $f : \Re^n \to \Re^n$ and $g : \Re^n \to \Re^{n\times m}$ are assumed to be continuously differentiable.

Let $h : \Re^n \to \Re$ be a continuously-differentiable function such that 0 is a regular value. Consider the set, $\mathcal{C} \in \Re^n$, given as the 0-super-level set of h is defined by:

$$\begin{aligned} \mathcal{C} &= \{x \in \mathcal{D} \subset \Re^n : h(x) \geq 0\} \\ \partial\mathcal{C} &= \{x \in \mathcal{D} \subset \Re^n : h(x) = 0\} \\ \mathrm{Int}(\mathcal{C}) &= \{x \in \mathcal{D} \subset \Re^n : h(x) > 0\}. \end{aligned} \tag{2}$$

Definition 1 [33]: Set $\mathcal{C}$ is forward invariant if for every $x_0 \in \mathcal{C}$, $x(t) \in \mathcal{C}$ for $x(0) = x_0$ and all $t \in I(x_0)$.

According to definition 1, the system is safe with respect to set $\mathcal{C}$ if $\mathcal{C}$ is forward invariant.

Definition 2 (Control Barrier Functions [33]): Let $\mathcal{C} \subset \mathcal{D} \subset \Re^n$ be the super-level set of a continuously differentiable function $h : \mathcal{D} \to \Re$, then h is a control barrier function (CBF) if there exists an extended class $\mathcal{K}_\infty$ function α such that for the system (1):

$$\sup_{u\in\Re^m} \{L_f h(x) + L_g h(x)u + \alpha(h(x))\} \geq 0, \tag{3}$$

where $L_f h(x) = \frac{\partial h(x)}{\partial x} f(x)$ and $L_g h(x) = \frac{\partial h(x)}{\partial x} g(x)$ are the lie derivatives of h along f and g, respectively.

B. Problem Formulation

Two reference frames are usually used for vessel modelling: an earth-fixed frame and a body-fixed frame. Consider a 3-DOF maneuvering model of an MASS. The kinematics and kinetics in the two reference frames are expressed as [34]:

$$\dot{\boldsymbol{\eta}} = \boldsymbol{R}(\psi)\boldsymbol{\nu} \tag{4}$$

and

$$\boldsymbol{M}\dot{\boldsymbol{\nu}} = -\boldsymbol{C}(\boldsymbol{\nu})\boldsymbol{\nu} - \boldsymbol{D}(\boldsymbol{\nu})\boldsymbol{\nu} + \boldsymbol{\tau} + \boldsymbol{w} \tag{5}$$

where $\boldsymbol{\eta} = [\boldsymbol{p}^T, \psi]^T \in \Re^3$ with $\boldsymbol{p} = [x, y]^T \in \Re^2$ denoting the position and $\psi \in (-\pi, \pi]$ denoting the heading in the earth-fixed inertial frame; $\boldsymbol{\nu} = [u, v, r]^T \in \Re^3$ denotes a vector of surge velocity, sway velocity, and yaw rate in the body-fixed frame; $\boldsymbol{\tau} = [\tau_1, \tau_2, \tau_3]^T \in \Re^3$ denotes a control input vector; $\boldsymbol{M}$ is a matrix composed of mass inertia and additional hydrodynamic inertia; $\boldsymbol{C}(\boldsymbol{\nu})$ is a matrix of Coriolis and centripetal terms; $\boldsymbol{D}(\boldsymbol{\nu})$ is a damping matrix; $\boldsymbol{w} = [w_1, w_2, w_3]^T \in \Re^3$ is a vector of disturbances caused by wind, waves, and ocean currents; $\boldsymbol{R}(\psi)$ is a rotation matrix defined as

$$\boldsymbol{R}(\psi) = \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{6}$$

Consider N_1 stationary obstacles O_o^s, $o = 1, ..., N_1$, which is modeled as a closed circular disk of radius h_o, centered at $\boldsymbol{p}_o = [x_o, y_o]^T \in \Re^2$. Denote $O_o^s = \{\boldsymbol{p} \in \Re^2 \,|\, \|\boldsymbol{p} - \boldsymbol{p}_o\| \leq h_o\}$. Next, consider N_2 moving obstacles O_j^d, $j = 1, ..., N_2$, which is modeled as a closed circular disk of radius h_j, centered at $\boldsymbol{p}_j = [x_j, y_j]^T \in \Re^2$. Denote $O_j^d = \{\boldsymbol{p} \in \Re^2 \,|\, \|\boldsymbol{p} - \boldsymbol{p}_j\| \leq h_j\}$.

Let the desired berthing position and heading angle for the MASS be defined as

$$\boldsymbol{\eta}_d = [x_d, y_d, \psi_d]^T, \tag{7}$$

where $x_d \in \Re$, $y_d \in \Re$ and $\psi_d \in \Re$.

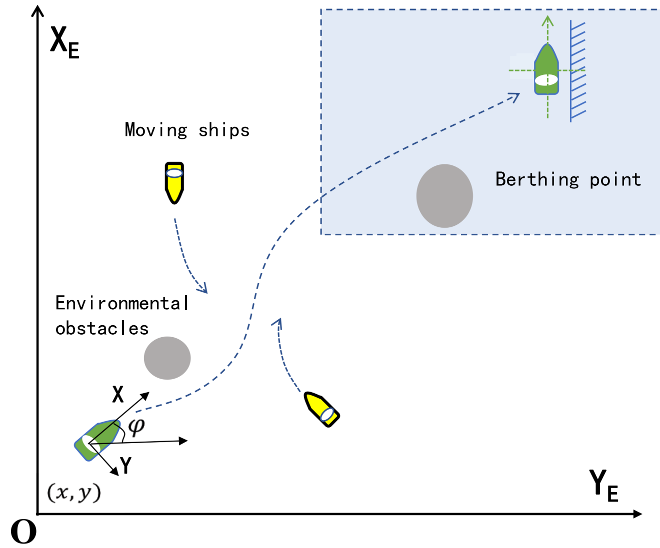

Fig. 1. An illustration of the automatic berthing of a maritime autonomous surface ship.

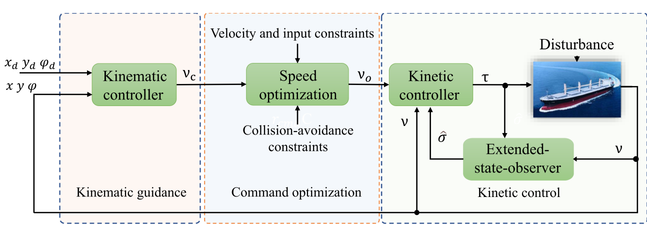

Fig. 2. A block diagram of the proposed control method for automatic berthing.

The control objective of automatic berthing is to design a safe control law for the MASS with the kinematics (4) and kinetics (5) such that the ship converges to the berthing position with a heading angle in (7), while avoiding collisions with stationary obstacles in O_o^s, moving obstacles O_j^s, and shorelines, as illustrated in Fig. 1.

III. Design and Analysis

Fig. 2 illustrates a block diagram of the control method for automatic berthing, including a kinematic control module, an anti-disturbance kinetic control module, and a command optimization module. The kinematic control module addresses the geometric guidance of an MASS with vehicle kinematics (4). The anti-disturbance kinetic control module handles vessel kinetics (5). The command optimization module optimizes control signals within various constraints.

A. Joint Guidance Law

During berthing, the MASS is supposed to align with a line parallel to the shoreline as follows :

$$a_0 x + b_0 y + c_0 = 0, \tag{8}$$

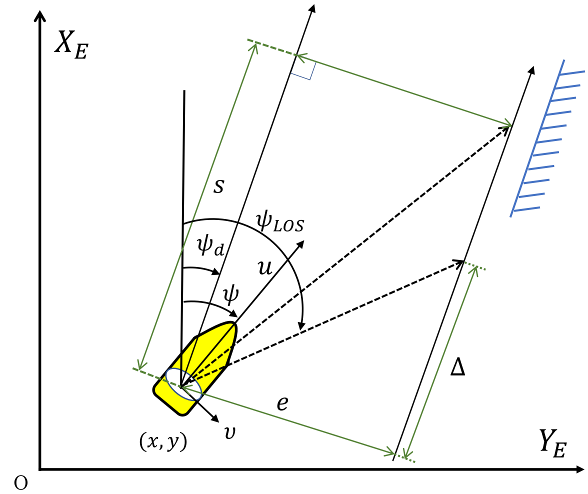

Fig. 3. An illustration of the line-of-sight guidance for automatic berthing.

where the berthing coordinates x_d and y_d satisfies

$$a_0 x_d + b_0 y_d + c_0 = 0, \tag{9}$$

and $a_0 \in \Re, b_0 \in \Re$, and $c_0 \in \Re$.

Then, the desired heading angle ψ_d is defined as

$$\psi_d = \operatorname{atan}\left(-\frac{a_0}{b_0}\right). \tag{10}$$

In order to achieve the automatic berthing, a position and heading tracking vector $\boldsymbol{z}_1 = \boldsymbol{\eta} - \boldsymbol{\eta}_d = [s, e, \psi_e]^T$ is defined as follows:

$$\boldsymbol{z}_1 = \begin{bmatrix} s \\ e \\ \psi_e \end{bmatrix} = \boldsymbol{R}^T(\psi_d) \begin{bmatrix} x - x_d \\ y - y_d \\ \psi - \psi_{LOS} \end{bmatrix} \tag{11}$$

where ψ_{LOS} is an LOS angle shown in Fig. 3 and defined as

$$\psi_{LOS} = \psi_d - \operatorname{atan}\left(\frac{e}{\Delta}\right) \tag{12}$$

where $\Delta > 0$ is a lookahead distance.

Letting $e_\psi = \psi - \psi_d$, the time derivative of z_1 along the kinematics (4) is given as

$$\begin{bmatrix} \dot{s} \\ \dot{e} \\ \dot{\psi}_e \end{bmatrix} = \begin{bmatrix} u\cos(e_\psi) - v\sin(e_\psi) \\ u\sin(e_\psi) + v\cos(e_\psi) \\ r - \dot{\psi}_{LOS} \end{bmatrix}$$

which leads to

$$\dot{\boldsymbol{z}}_{\boldsymbol{1}} = \boldsymbol{R}(\boldsymbol{e_\psi})\boldsymbol{\nu} - \left[0, 0, \dot{\psi}_{LOS}\right]^T. \tag{13}$$

To achieve point-heading stabilization, a velocity and yaw rate guidance law $\boldsymbol{\nu_c} = [u_c, v_c, r_c]^T$ is proposed based on the LOS guidance as follows:

$$\begin{bmatrix} u_c \\ v_c \\ r_c \end{bmatrix} = \boldsymbol{R^T}(\boldsymbol{e_\psi}) \begin{bmatrix} -\frac{k_1 s}{\sqrt{s^2+\Delta_1^2}} \\ -\frac{k_2 e}{\sqrt{e^2+\Delta_2^2}} \\ -\frac{k_3 \psi_e}{\sqrt{\psi_e^2+\Delta_3^2}} + \dot{\psi}_{LOS} \end{bmatrix}, \tag{14}$$

where $k_1 \in \Re$, $k_2 \in \Re$, $k_3 \in \Re$, $\Delta_1 \in \Re$, $\Delta_2 \in \Re$, and $\Delta_3 \in \Re$ are control constants to be designed.

Letting $\boldsymbol{\nu}_e = [u_e, v_e, r_e]^T$ with $u_e = u - u_c$, $v_e = v - v_c$, and $r_e = r - r_c$, the LOS-based guidance law (14) results in the following closed-loop guidance system expressed as

$$\dot{\boldsymbol{z}}_1 = -\boldsymbol{K}_g \boldsymbol{z}_1 + \boldsymbol{\nu}_e, \tag{15}$$

where $\boldsymbol{K}_g = \mathrm{diag}\{k_1/\sqrt{s^2+\Delta_1^2}, k_2/\sqrt{e^2+\Delta_2^2}, k_3/\sqrt{\psi_e^2+\Delta_3^2}\}$.

Lemma 1: The closed-loop error kinetics (15) with the state vector being $\boldsymbol{z_1}$ and the input vector being $\boldsymbol{\nu_e}$ is input-to-state stable.

Proof: Consider a Lyapunov function

$$V_1 = \sqrt{\frac{1}{2}\boldsymbol{z}_1^T \boldsymbol{z}_1}. \tag{16}$$

Taking the time derivative of V along (15) yields

$$\begin{aligned} \dot{V}_1 &= \frac{-\boldsymbol{z}_1^T \boldsymbol{K}_g \boldsymbol{z}_1 + \boldsymbol{z}_1^T \boldsymbol{\nu}_e}{2V_1} \\ &\leq -\alpha_1 V_1 + \frac{\|\boldsymbol{\nu}_e\|}{\sqrt{2}}. \end{aligned} \tag{17}$$

where $\alpha_1 = \lambda_{\min}(\boldsymbol{K}_g)$ for $\|\boldsymbol{z}_1\| \leq \boldsymbol{M} < \infty$. It is concluded that subsystem (15) is input-to-state stable. ∎

B. Control Law

This section presents the control law for the MASS kinetics (5). To facilitate the ensuing derivations, the kinetic (5) is rewritten as

$$\dot{\boldsymbol{\nu}} = \boldsymbol{\sigma} + (\boldsymbol{M}^*)^{-1}\boldsymbol{\tau}, \tag{18}$$

where $\boldsymbol{M}^*$ is a nominal inertia parameter for $\boldsymbol{M}$ and $\boldsymbol{\sigma}$ is a total disturbance expressed as

$$\begin{aligned} \boldsymbol{\sigma} &= \boldsymbol{M}^{-1}(-\boldsymbol{C}(\boldsymbol{\nu})\boldsymbol{\nu} - \boldsymbol{D}(\boldsymbol{\nu})\boldsymbol{\nu} + \boldsymbol{w}) + (\boldsymbol{M}^{-1} \\ &\quad - (\boldsymbol{M}^*)^{-1})\boldsymbol{\tau}. \end{aligned} \tag{19}$$

Similar to [35], [36], $\dot{\boldsymbol{\sigma}}$ is assumed to be bounded. This assumption is reasonable because the energy of external disturbance is limited. Moreover, the motion of the ship is subject to hydrodynamic damping and its velocities and accelerations are reasonably bounded.

Let $\hat{\boldsymbol{\sigma}}$ be the estimation of $\boldsymbol{\sigma}$, and an extended-state-observer (ESO) is developed as:

$$\begin{cases} \dot{\hat{\boldsymbol{\nu}}} = \hat{\boldsymbol{\sigma}} + (\boldsymbol{M}^*)^{-1}\boldsymbol{\tau} - \boldsymbol{K}_1(\hat{\boldsymbol{\nu}} - \boldsymbol{\nu}) \\ \dot{\hat{\boldsymbol{\sigma}}} = -\boldsymbol{K}_2(\hat{\boldsymbol{\nu}} - \boldsymbol{\nu}) \end{cases} \tag{20}$$

where $[\boldsymbol{K}_1, \boldsymbol{K}_2]^T = [2\omega_o \boldsymbol{I}_3, \omega_o^2 \boldsymbol{I}_3]^T$ where $\boldsymbol{\omega_o} > 0$ is a desired observer bandwidth. Letting $\tilde{\boldsymbol{\nu}} = \hat{\boldsymbol{\nu}} - \boldsymbol{\nu}$ and $\tilde{\boldsymbol{\sigma}} = \hat{\boldsymbol{\sigma}} - \boldsymbol{\sigma}$, the resulting error kinetics of $\boldsymbol{z}_2 = [\tilde{\boldsymbol{\nu}}^T, \tilde{\boldsymbol{\sigma}}^T]^T$ is expressed as

$$\dot{\boldsymbol{z}}_2 = \boldsymbol{A}\boldsymbol{z}_2 - \boldsymbol{B}\dot{\boldsymbol{\sigma}} \tag{21}$$

where

$$\boldsymbol{A} = \begin{bmatrix} -\boldsymbol{K}_1 & \boldsymbol{I}_3 \\ -\boldsymbol{K}_2 & \boldsymbol{0} \end{bmatrix}, \boldsymbol{B} = \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{I}_3 \end{bmatrix}. \tag{22}$$

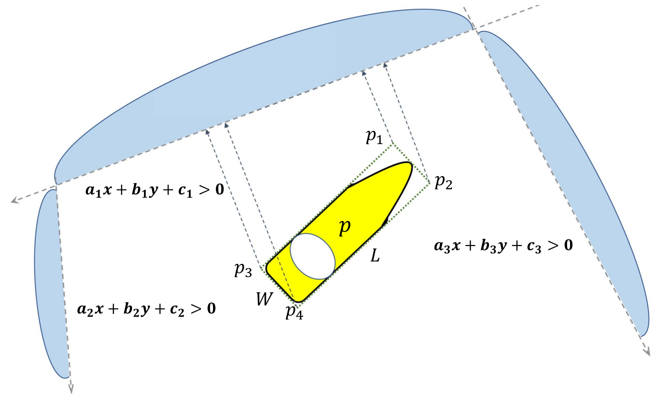

Fig. 4. An illustration of the shoreline constraints.

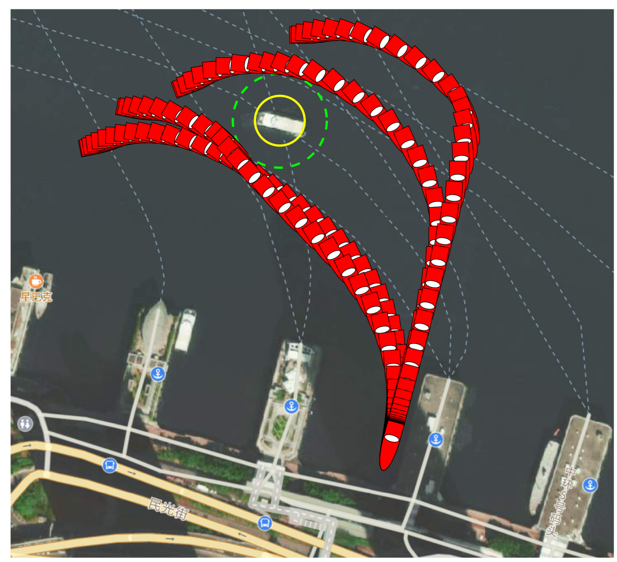

Fig. 5. The trajectories of the vessel with the proposed safety-aware automatic berthing control law from four starting points and angles.

As $\boldsymbol{A}$ is Hurwitz, there exists a positive-definite matrix $\boldsymbol{P}$ such that

$$\boldsymbol{P}\boldsymbol{A} + \boldsymbol{A}^T\boldsymbol{P} + \varepsilon \boldsymbol{I}_6 = 0 \tag{23}$$

where ε is a positive constant.

Let $\boldsymbol{\nu}_s \in \Re^3$ be a safe velocity vector and $\boldsymbol{\nu}_e = \boldsymbol{\nu} - \boldsymbol{\nu}_s \in \Re^3$ be a velocity-tracking error vector. By using the estimated information $\hat{\boldsymbol{\sigma}}$ from the ESO (20), an anti-disturbance kinetic control law is proposed as:

$$\boldsymbol{\tau} = \boldsymbol{M}^*(-\boldsymbol{K}_c\boldsymbol{\nu}_e - \hat{\boldsymbol{\sigma}}) \tag{24}$$

where $\boldsymbol{K}_c \in \Re^{3\times 3}$ is a control gain matrix.

The control law in (24) results in the following closed-loop kinetic control system

$$\dot{\boldsymbol{\nu}}_e = -\boldsymbol{K}_c\boldsymbol{\nu}_e - \tilde{\boldsymbol{\sigma}} - \dot{\boldsymbol{\nu}}_s. \tag{25}$$

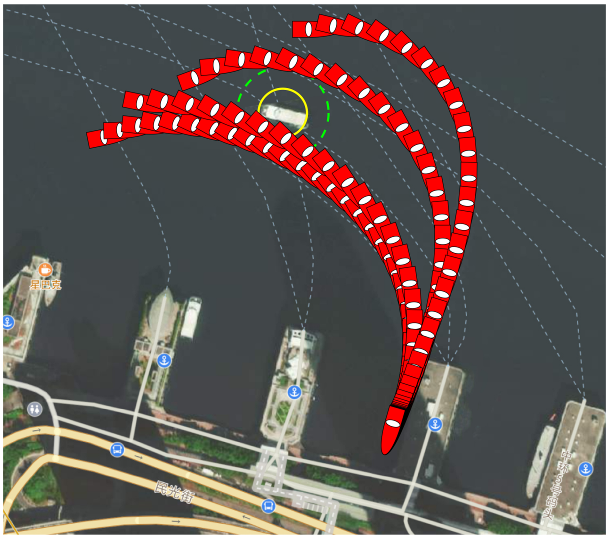

Fig. 6. The trajectories of the vessel with the automatic berthing control law in (54) from four starting points and angles.

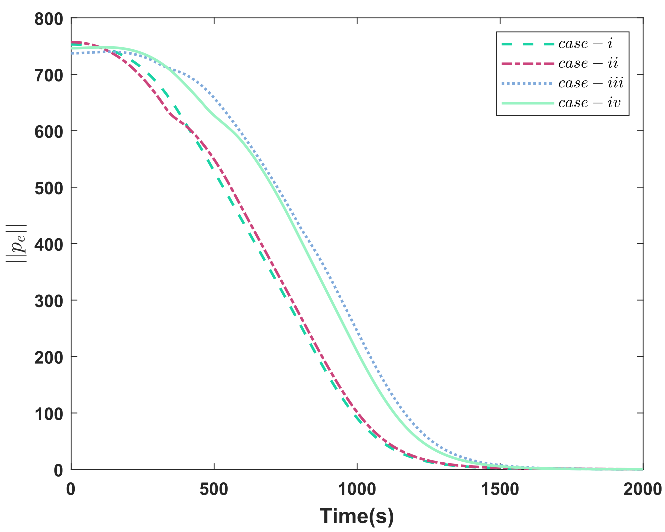

Fig. 7. The norms of the position tracking errors $p_e = [s, e]^T$ during automatic berthing.

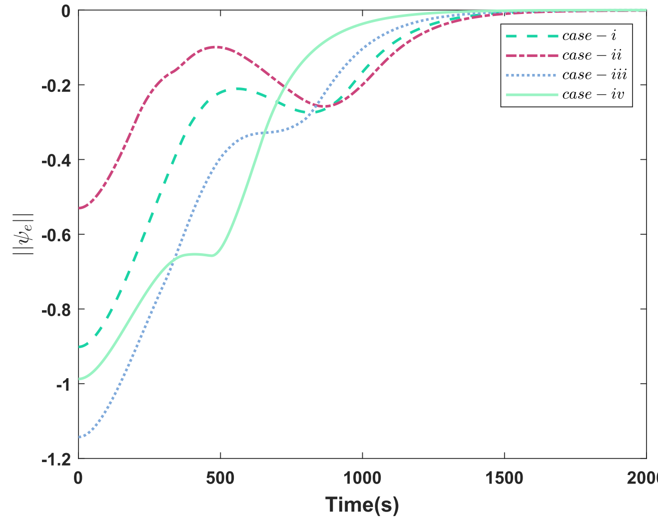

Fig. 8. The norms of the heading tracking errors during automatic berthing.

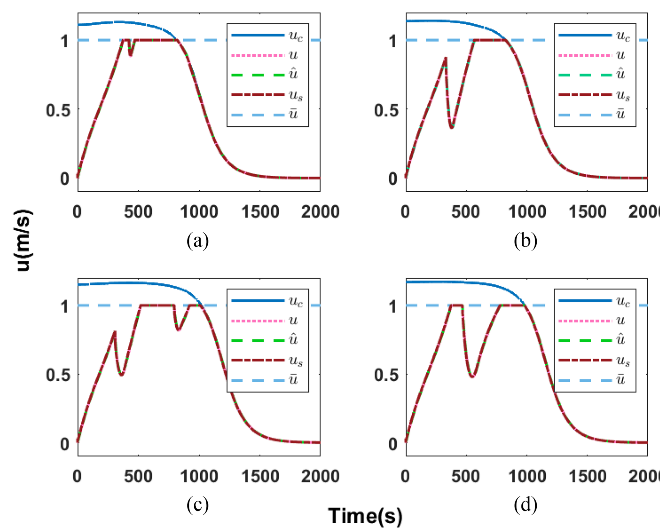

Fig. 9. Commanded signal, actual signal, estimated signal, and optimized signal of the surge velocity. (a) Case i, (b) Case ii, (c) Case iii, (d) Case iv.

Lemma 2: The closed-loop kinetic control system, consisting of subsystems (21) and (25) with the state vectors being $\boldsymbol{z}_2$ and $\boldsymbol{\nu}_e$ and the input vectors being $\dot{\boldsymbol{\sigma}}$ and $\dot{\boldsymbol{\nu}}_s$, is input-to-state stable.

Proof: Consider a Lyapunov function

$$V_2 = \sqrt{\frac{1}{2}\boldsymbol{z}_2^T \boldsymbol{P} \boldsymbol{z}_2} + \sqrt{\frac{1}{2}\boldsymbol{\nu}_e^T \boldsymbol{\nu}_e}, \tag{26}$$

and its time derivative along (21) is given by

$$\begin{aligned}\dot{V}_2 \leq & \frac{1}{\sqrt{2\boldsymbol{z}_2^T \boldsymbol{P} \boldsymbol{z}_2}} \left(-\varepsilon\|\boldsymbol{z}_2\|^2 + 2\|\boldsymbol{z}_2\|\|\|\boldsymbol{PB}\|\|\dot{\boldsymbol{\sigma}}\|\right) \\ & + \frac{1}{\sqrt{2\boldsymbol{\nu}_e^T \boldsymbol{\nu}_e}} \left(-\lambda_{\min}(\boldsymbol{K}_c)\|\boldsymbol{\nu}_e\|^2 + \|\boldsymbol{\nu}_e\|(\|\tilde{\boldsymbol{\sigma}}\| + \|\dot{\boldsymbol{\nu}}_s\|)\right), \\ \leq & -\frac{\varepsilon}{2\lambda_{\max}(\boldsymbol{P})}\sqrt{\frac{1}{2}\boldsymbol{z}_2^T \boldsymbol{P} \boldsymbol{z}_2} + \frac{\|\boldsymbol{PB}\|\|\dot{\boldsymbol{\sigma}}\|}{\sqrt{2\lambda_{\min}(\boldsymbol{P})}}, \\ & -\lambda_{\min}(\boldsymbol{K}_c)\sqrt{\frac{1}{2}\boldsymbol{\nu}_e^T \boldsymbol{\nu}_e} + (\|\tilde{\boldsymbol{\sigma}}\| + \|\dot{\boldsymbol{\nu}}_s\|)\frac{1}{\sqrt{2}}. \\ \leq & -\left(\frac{\varepsilon}{2\lambda_{\max}(\boldsymbol{P})} - \frac{1}{\sqrt{\lambda_{\min}(\boldsymbol{P})}}\right)\sqrt{\frac{1}{2}\boldsymbol{z}_2^T \boldsymbol{P} \boldsymbol{z}_2} \\ & -\lambda_{\min}(\boldsymbol{K}_c)\sqrt{\frac{1}{2}\boldsymbol{\nu}_e^T \boldsymbol{\nu}_e} + \frac{\|\boldsymbol{PB}\|\|\dot{\boldsymbol{\sigma}}\|}{\sqrt{2\lambda_{\min}(\boldsymbol{P})}} + \frac{\|\dot{\boldsymbol{\nu}}_s\|}{\sqrt{2}}. \end{aligned} \tag{27}$$

By selecting $\min\{\varepsilon/2\lambda_{\max}(\boldsymbol{P}) - 1/\sqrt{\lambda_{\min}(\boldsymbol{P})}, \lambda_{\min}(\boldsymbol{K}_c)\} = \alpha_2 > 0$, it follows that

$$\dot{V}_2 \leq -\alpha_2 V_2 + \frac{\|\boldsymbol{PB}\|\|\dot{\boldsymbol{\sigma}}\|}{\sqrt{2\lambda_{\min}(\boldsymbol{P})}} + \frac{\|\dot{\boldsymbol{\nu}}_s\|}{\sqrt{2}}. \tag{28}$$

It follows that subsystems (21) and (25) are input-to-state stable. ■

C. Command Optimization

In this subsection, we proceed to present a command optimization method for generating the safe guidance vector $\boldsymbol{\nu_s}$ satisfying the velocity, input, obstacle-avoidance constraints, by modifying the nominal guidance vector $\boldsymbol{\nu_c}$ in a minimally invasive fashion.

1) Velocity and Input Constraints: To satisfy the velocity and yaw rate constraints, the following constraints on the desired guidance vector $\boldsymbol{\nu_s}$ are to be enforced:

$$\underline{\boldsymbol{\nu}} \leq \boldsymbol{\nu}_s \leq \bar{\boldsymbol{\nu}}, \tag{29}$$

where $\underline{\boldsymbol{\nu}} = [\underline{u}, \underline{v}, \underline{r}]$, $\bar{\boldsymbol{\nu}} = [\bar{u}, \bar{v}, \bar{r}]$ with $\underline{u} \in \Re$, $\bar{u} \in \Re$, $\underline{v} \in \Re$, $\bar{v} \in \Re$, $\underline{r} \in \Re$, and $\bar{r} \in \Re$ are designed to satisfy $u_{\min} < \underline{u}$, $\bar{u} < u_{\max}$, $v_{\min} < \underline{v}$, $\bar{v} < v_{\max}$, $r_{\min} < \underline{r}$, and $\bar{r} < r_{\max}$.

In view of the limited driving forces of the MASS, the control input vector $\boldsymbol{\tau}$ is bounded by

$$\underline{\boldsymbol{\tau}} \leq \boldsymbol{\tau} \leq \bar{\boldsymbol{\tau}}, \tag{30}$$

where $\underline{\boldsymbol{\tau}} = [\underline{\tau}_1, \underline{\tau}_2, \underline{\tau}_3]^T$ and $\bar{\boldsymbol{\tau}} = [\bar{\tau}_1, \bar{\tau}_2, \bar{\tau}_3]^T$ with $\underline{\tau}_1 \in \Re$, $\underline{\tau}_2 \in \Re$, $\underline{\tau}_3 \in \Re$, $\bar{\tau}_1 \in \Re$, $\bar{\tau}_2 \in \Re$, and $\bar{\tau}_3 \in \Re$ being constants.

Substituting (30) into (24) yields the following constraints on the velocity and yaw rate vector ν_s:

$$\begin{cases} \boldsymbol{\nu}_s \leq \boldsymbol{\nu} + \boldsymbol{K}_c^{-1}\hat{\boldsymbol{\sigma}} + \boldsymbol{K}_c^{-1}(\boldsymbol{M}^*)^{-1}\bar{\boldsymbol{\tau}} \\ -\boldsymbol{\nu}_s \leq -\boldsymbol{\nu} - \boldsymbol{K}_c^{-1}\hat{\boldsymbol{\sigma}} - \boldsymbol{K}_c^{-1}(\boldsymbol{M}^*)^{-1}\underline{\boldsymbol{\tau}}. \end{cases} \tag{31}$$

2) Collision-Avoidance Constraint for Stationary Obstacles: Let $\boldsymbol{p}_o = [x_o, y_o]^T$ be the position of a stationary obstacle. To avoid colliding with stationary obstacles, the following collision-avoidance constraint is encoded in a control barrier function:

$$h_o(\boldsymbol{p}, \boldsymbol{p}_o) = \|\boldsymbol{p} - \boldsymbol{p}_o\| - \underline{d}_o, \tag{32}$$

where $\underline{d}_o$ denotes the safe distance between the MASS and the obstacle and $h_o(\boldsymbol{p}, \boldsymbol{p}_o)$ satisfies $\|\partial h_o(\boldsymbol{p}, \boldsymbol{p}_o)/\partial \boldsymbol{p}\| \leq C_{ho}$ with C_{ho} being a positive constant.

Letting $\boldsymbol{q} = [u, v]^T$, it follows from (4) that

$$\dot{\boldsymbol{p}} = \underline{\boldsymbol{R}}(\psi)\boldsymbol{q}, \tag{33}$$

where the matrix $\underline{\boldsymbol{R}}(\psi)$ is a 2×2 submatrix of $\boldsymbol{R}(\psi)$ and

$$\underline{\boldsymbol{R}}(\psi) = \begin{bmatrix} \cos(\psi) & -\sin(\psi) \\ \sin(\psi) & \cos(\psi) \end{bmatrix}. \tag{34}$$

Ensuring that

$$\frac{\partial h_o(\boldsymbol{p}, \boldsymbol{p}_o)}{\partial \boldsymbol{p}} \dot{\boldsymbol{p}} = \frac{(\boldsymbol{p} - \boldsymbol{p}_o)^T}{\|\boldsymbol{p} - \boldsymbol{p}_o\|} \underline{\boldsymbol{R}}(\psi)\boldsymbol{q} \geq -\gamma_o h_o(\boldsymbol{p}, \boldsymbol{p}_o) \tag{35}$$

guarantees that the stationary obstacle can be avoided, where $\gamma_o \in \Re$ is a positive constant.

3) Collision-Avoidance Constraint for Moving Obstacles: Let $\boldsymbol{p}_j = [x_j, y_j]^T$ be the position of a moving obstacle. To avoid colliding with the jth moving obstacle, the following collision-avoidance constraint is encoded in a control barrier function:

$$h_j(\boldsymbol{p}, \boldsymbol{p}_j) = \|\boldsymbol{p} - \boldsymbol{p}_j\| - \underline{d}_j, \tag{36}$$

where $\underline{d}_j$ denotes the safe distance between the MASS and the obstacle and $h_j(\boldsymbol{p}, \boldsymbol{p}_j)$ satisfies $\|\partial h_j(\boldsymbol{p}, \boldsymbol{p}_j)/\partial \boldsymbol{p}\| \leq C_{hj}$ with C_{hj} being a positive constant.The inequality

$$\dot{h}_j(\boldsymbol{p}, \boldsymbol{p}_j) \geq -\gamma_j h_j(\boldsymbol{p}, \boldsymbol{p}_j) \tag{37}$$

ensures that the MASS can avoid the moving obstacle with $\gamma_j \in \Re$ being a positive constant, and gives

$$\frac{(\boldsymbol{p} - \boldsymbol{p}_j)^T}{\|\boldsymbol{p} - \boldsymbol{p}_j\|} \left(\underline{\boldsymbol{R}}(\psi)\boldsymbol{q} - \dot{\boldsymbol{p}}_j\right) \geq -\gamma_j h_j(\boldsymbol{p}, \boldsymbol{p}_j). \tag{38}$$

4) Collision-Avoidance Constraint for the Shoreline: Assume that the shorelines can be expressed with the following linear equations:

$$a_k x + b_k y + c_k = 0, \ k = 1, ..., n, \tag{39}$$

where $a_k \in \Re$, $b_k \in \Re$, and $c_k \in \Re$. It is assumed that the shape of the ship is approximated by a rectangle of length L and width W and the four boundary points can be expressed

$$\begin{aligned} \boldsymbol{p}_1 &= \boldsymbol{p} + \underline{\boldsymbol{R}}(\psi)[L/2, W/2]^T, \\ \boldsymbol{p}_2 &= \boldsymbol{p} + \underline{\boldsymbol{R}}(\psi)[L/2, -W/2]^T, \\ \boldsymbol{p}_3 &= \boldsymbol{p} + \underline{\boldsymbol{R}}(\psi)[-L/2, W/2]^T, \\ \boldsymbol{p}_4 &= \boldsymbol{p} + \underline{\boldsymbol{R}}(\psi)[-L/2, -W/2]^T. \end{aligned} \tag{40}$$

The collision avoidance with the kth shoreline is encoded in the following control barrier function:

$$h_c(\boldsymbol{p}, \boldsymbol{p}_i) = \frac{a_k x_i + b_k y_i + c_k}{\sqrt{a_k^2 + b_k^2}} - \underline{d}_k, \tag{41}$$

where $i = 1, 2, 3, 4$, $k = 1, ..., n$, $\underline{d}_k \in \Re$ is a positive constant, and $h_c(\boldsymbol{p}, \boldsymbol{p}_i)$ satisfies $\|\partial h_c(\boldsymbol{p}, \boldsymbol{p}_i)/\partial \boldsymbol{p}\| \leq C_{hc}$ with C_{hc} being a positive constant.

The following inequality

$$\dot{h}_c(\boldsymbol{p}, \boldsymbol{p}_i) \geq -\gamma_c h_c(\boldsymbol{p}, \boldsymbol{p}_i), \tag{42}$$

enforces the shoreline constraint satisfaction, where $\gamma_c \in \Re$ is a positive constant. As a result, the following constraint is imposed on the velocity vector as follows:

$$\begin{aligned} \frac{\partial h_c(\boldsymbol{p}_i)}{\partial \boldsymbol{p}} \dot{\boldsymbol{p}} &= \left[\frac{a_k}{\sqrt{a_k^2 + b_k^2}}, \frac{b_k}{\sqrt{a_k^2 + b_k^2}}\right] \underline{\boldsymbol{R}}(\psi)\boldsymbol{q} \\ &\geq -\gamma_c h_c(\boldsymbol{p}, \boldsymbol{p}_i). \end{aligned} \tag{43}$$

5) Quadratic Optimization: Note that the collision-avoidance constraints are imposed on the velocity vector q, which has to be achieved by using the low-level kinetic control law. Toward it, the collision avoidance constraints are expressed as

$$\begin{cases} \boldsymbol{A}_o \boldsymbol{q}_s \leq \gamma_o h_o(\boldsymbol{p}, \boldsymbol{p}_o), \\ \boldsymbol{A}_j \boldsymbol{q}_s \leq \gamma_j h_j(\boldsymbol{p}, \boldsymbol{p}_j) - 2(\boldsymbol{p} - \boldsymbol{p}_j)^T \dot{\boldsymbol{p}}_j, \\ \boldsymbol{A}_k \boldsymbol{q}_s \leq \gamma_c h_c(\boldsymbol{p}, \boldsymbol{p}_i), \end{cases} \tag{44}$$

TABLE I
THE CONTROL PARAMETERS IN THE SIMULATIONS

parameter	value	unit	parameter	value	unit
a_0	-3.56	-	k_1	1.2	-
b_0	-1	-	k_2	0.35	-
c_0	3540	-	k_3	0.15	-
Δ_1	150	-	γ_o	0.3	-
Δ_2	150	-	γ_j	0.3	-
Δ_3	20	-	γ_c	0.18	-
K_{c1}	5	-	K_{c2}	5	-
K_{c3}	4	-	ρ_a	1.224	-
μ	0.0	-	κ	100	-
$\underline{u}$	-0.3	m/s	$\bar{u}$	1	m/s
$\underline{v}$	-0.25	m/s	$\bar{v}$	0.25	m/s
$\underline{r}$	-0.05	rad/s	$\bar{r}$	0.05	rad/s
$\underline{\tau}_1$	-10^6	N·m	$\bar{\tau}_1$	10^6	N·m
$\underline{\tau}_2$	-10^6	N·m	$\bar{\tau}_2$	10^6	N·m
$\underline{\tau}_3$	-10^7	N·m	$\bar{\tau}_3$	10^7	N·m
p_{o1}	552	m	p_{o2}	205	m
$\underline{d}_o$	110	m	$\underline{d}_j$	100	m
$\underline{d}_k$	10	m	A_{Fw}	300	m^2
A_{Lw}	900	m^2	L_{oa}	100	m
ω_o	10	-			

TABLE II
STARTING POINT POSITION IN CASES I-IV

	x(m)	y(m)	ψ(rad)
Case i	220	250	-0.2
Case ii	287	185	0.2
Case iii	389	140	0.35
Case iv	600	50	0.0

TABLE III
MINIMAL DISTANCES RELATIVE TO THE OBSTACLE AND SHORELINE

	relative to the obstacle		relative to the shoreline	
	with constraints	without constraints	with constraints	without constraints
Case i	112.5 m	70.58 m	20.6 m	12.4 m
Case ii	111.1 m	48.71 m	20.6 m	10.6 m
Case iii	110.1 m	95.58 m	20.3 m	0.75 m
Case iv	162.3 m	133.9 m	20.1 m	-0.8 m

where

$$\begin{cases} \boldsymbol{A}_o = -(\boldsymbol{p}-\boldsymbol{p}_o)^T \underline{\boldsymbol{R}}(\psi)/\|\boldsymbol{p}-\boldsymbol{p}_o\| \\ \boldsymbol{A}_j = (\boldsymbol{p}-\boldsymbol{p}_j)^T \underline{\boldsymbol{R}}(\psi)/\|\boldsymbol{p}-\boldsymbol{p}_j\| \\ \boldsymbol{A}_k = \left[-a_k/\sqrt{a_k^2+b_k^2}, -b_k/\sqrt{a_k^2+b_k^2}\right] \underline{\boldsymbol{R}}(\psi). \end{cases} \tag{45}$$

As such, the safe guidance signal vector $\boldsymbol{\nu_s}$ is obtained by solving the following quadratic program as follows

$$\begin{aligned} &\min_{\boldsymbol{\nu}_s\in\Re^3} \|\boldsymbol{\nu}_s-\boldsymbol{\nu}_c\|^2 \\ &\text{s.t. } \boldsymbol{\nu}_s \le \boldsymbol{\nu} + \boldsymbol{K}^{-1}\hat{\boldsymbol{\sigma}} + \boldsymbol{K}^{-1}(\boldsymbol{M}^*)^{-1}\bar{\boldsymbol{\tau}}, \\ &\quad -\boldsymbol{\nu}_s \le -\boldsymbol{\nu} - \boldsymbol{K}^{-1}\hat{\boldsymbol{\sigma}} - \boldsymbol{K}^{-1}(\boldsymbol{M}^*)^{-1}\underline{\boldsymbol{\tau}}, \\ &\quad \underline{\boldsymbol{\nu}} \le \boldsymbol{\nu}_s \le \bar{\boldsymbol{\nu}}, \\ &\quad \boldsymbol{A}_o\boldsymbol{q}_s \le \gamma_o h_o(\boldsymbol{p},\boldsymbol{p}_o), \\ &\quad \boldsymbol{A}_j\boldsymbol{q}_s \le \gamma_j h_j(\boldsymbol{p},\boldsymbol{p}_j) - 2(\boldsymbol{p}-\boldsymbol{p}_j)^T\dot{\boldsymbol{p}}_j, \\ &\quad \boldsymbol{A}_k\boldsymbol{q}_s \le \gamma_c h_c(\boldsymbol{p},\boldsymbol{p}_i). \end{aligned} \tag{46}$$

Generating the real-time control signals for the automatic berthing an MASS entails solving problem (46) efficiently. As an efficient optimization solver, a neurodynamic optimization method is employed herein to solve problem (46). A projection neural network in [37] is customized as follows:

$$\begin{bmatrix} \dot{\boldsymbol{\nu}}_s \\ \dot{\boldsymbol{\xi}} \end{bmatrix} = \kappa \begin{bmatrix} -\boldsymbol{\nu}_s + P_\nu(\boldsymbol{\nu}_s - (\nabla J(\boldsymbol{\nu}_s) + \nabla\boldsymbol{\zeta}(\boldsymbol{\nu}_s)\boldsymbol{\xi})) \\ -\boldsymbol{\xi} + (\boldsymbol{\xi} + \boldsymbol{\zeta}(\boldsymbol{\nu}_s))^+ \end{bmatrix}, \tag{47}$$

where κ is a positive scaling constant, $J(\boldsymbol{\nu}_s) = \|\boldsymbol{\nu}_s-\boldsymbol{\nu}_c\|^2, \boldsymbol{\xi} \in \Re^6$, $(\boldsymbol{\xi})^+ = [(\boldsymbol{\xi}_1)^+,\ldots,(\boldsymbol{\xi}_m)^+]^T$ with $(\boldsymbol{\xi}_1)^+ = \max\{0,\boldsymbol{\xi}_1\}$ and $P_{\boldsymbol{\nu}}(\cdot)$ are projection operators (piecewise activation functions), and $\boldsymbol{\zeta}(\boldsymbol{\nu}_s)$: $\Re^3 \to \Re^6$ is defined as:

$$\boldsymbol{\zeta}(\boldsymbol{\nu}_s) = \begin{bmatrix} \boldsymbol{A}_o\boldsymbol{q}_s - \gamma_o h_o(\boldsymbol{p},\boldsymbol{p}_o) \\ \boldsymbol{A}_j\boldsymbol{q}_s - \gamma_j h_j(\boldsymbol{p},\boldsymbol{p}_j) - 2(\boldsymbol{p}-\boldsymbol{p}_j)^T\dot{\boldsymbol{p}}_j \\ \boldsymbol{A}_k\boldsymbol{q}_s - \gamma_c h_c(\boldsymbol{p},\boldsymbol{p}_i) \end{bmatrix}.$$

The piecewise-linear activation function (projection operator) $P_\nu(\boldsymbol{\nu}) : \Re^3 \to \Re^3$ is defined as $P_\nu(\boldsymbol{\nu}) = \min\{\max\{\boldsymbol{\nu}_{\min},\boldsymbol{\nu}\},\boldsymbol{\nu}_{\max}\}$, where $\boldsymbol{\nu}_{\min} = \max\{\boldsymbol{\nu}_2,\underline{\boldsymbol{\nu}}\}$, and $\boldsymbol{\nu}_{\max} = \min\{\boldsymbol{\nu}_1,\bar{\boldsymbol{\nu}}\}$ with $\boldsymbol{\nu}_1 = \boldsymbol{\nu} + \boldsymbol{K}^{-1}\hat{\boldsymbol{\sigma}} + \boldsymbol{K}^{-1}(\boldsymbol{M}^*)^{-1}\bar{\boldsymbol{\tau}}$ and $\boldsymbol{\nu}_2 = \boldsymbol{\nu} + \boldsymbol{K}^{-1}\hat{\boldsymbol{\sigma}} + \boldsymbol{K}^{-1}(\boldsymbol{M}^*)^{-1}\underline{\boldsymbol{\tau}}$.

With any initial point $(\boldsymbol{\nu}_s(t_0),\boldsymbol{\xi}(t_0))$ satisfying $\underline{\boldsymbol{\nu}} \le \boldsymbol{\nu}_s(t_0) \le \bar{\boldsymbol{\nu}}$ and $\boldsymbol{\nu}_s(t_0) \ge 0$, the output of projection neural network (47) is exponentially convergent to $\boldsymbol{\nu}_s^*$ [38], where $\boldsymbol{\nu}_s^*$ is the optimal solution to problem (46).

The scaling parameter κ in (47) determines the rate of convergence. By selecting a large value of κ, the safety-critical control signal can be obtained in real time. Theoretically, the scaling parameter κ with a large value ensures a high convergence rate of continuous-time projection neural network (47). However, in practice, the continuous-time neural network is implemented by a discrete-time system with sufficiently small stepsizes, and the value of the scaling parameter κ depends on the stepsizes. The scaling parameter κ with an excessively large value may result in overshooting and a long simulation time. As such, a trade-off should be made.

To facilitate the ensuing input-to-state safety analysis, a set Ω is defined as follows:

$$\begin{aligned} \Omega = \{p \in C : h_o(\boldsymbol{p},\boldsymbol{p}_o) \ge 0, h_j(\boldsymbol{p},\boldsymbol{p}_j) \ge 0, \\ h_c(\boldsymbol{p},\boldsymbol{p}_i) \ge 0\}. \end{aligned} \tag{48}$$

Due to the bounded disturbance $\dot{\boldsymbol{\sigma}}$ and $\dot{\boldsymbol{\nu}}_s$, the asymptotical tracking of the safe velocity vector $\boldsymbol{\nu}_s$ is not possible.

As a consequence, the proposed control law may achieve input-to-state safety only instead of safety; i.e., the invariance of the larger set $\Omega_s \subseteq \Omega$ as

$$\begin{aligned} \Omega_s = \{\boldsymbol{p} \in C : \bar{h}_o(\boldsymbol{p},\boldsymbol{p}_o) \ge 0, \bar{h}_j(\boldsymbol{p},\boldsymbol{p}_j) \ge 0, \\ \bar{h}_c(\boldsymbol{p},\boldsymbol{p}_i) \ge 0\} \end{aligned} \tag{49}$$

where $\bar{h}_o(\boldsymbol{p},\boldsymbol{p}_o) = h_o(\boldsymbol{p},\boldsymbol{p}_o) + \iota_o(\|\epsilon\|_\infty)$, $\bar{h}_j(\boldsymbol{p},\boldsymbol{p}_j) = h_j(\boldsymbol{p},\boldsymbol{p}_j) + \iota_j(\|\epsilon\|_\infty)$, and $\bar{h}_c(\boldsymbol{p},\boldsymbol{p}_i) = h_c(\boldsymbol{p},\boldsymbol{p}_i) + \iota_c(\|\epsilon\|_\infty)$ with ι_o, ι_j, ι_c being class $\mathcal{K}$ functions to be specified and $\epsilon = [\|\dot{\boldsymbol{\sigma}}\|,\|\dot{\boldsymbol{\nu}}_s\|]^T$.

A kinetic extension can be obtained as follows:

$$\Omega_v = \{\boldsymbol{p} \in C : \hbar_o(\boldsymbol{p}, \boldsymbol{p}_o) \geq 0, \hbar_j(\boldsymbol{p}, \boldsymbol{p}_j) \geq 0, \hbar_c(\boldsymbol{p}, \boldsymbol{p}_i) \geq 0\}; \tag{50}$$

where $\hbar_o(\boldsymbol{p}, \boldsymbol{p}_o) = -V_2(\boldsymbol{\nu}_e, \boldsymbol{z}_2) + \rho_o h_o(\boldsymbol{p}, \boldsymbol{p}_o) + \rho_o \iota_o(\|\epsilon\|_\infty)$, $\hbar_j(\boldsymbol{p}, \boldsymbol{p}_j) = -V_2(\boldsymbol{\nu}_e, \boldsymbol{z}_2) + \rho_j h_j(\boldsymbol{p}, \boldsymbol{p}_j) + \rho_j \iota_j(\|\epsilon\|_\infty)$, and $\hbar_c(\boldsymbol{p}, \boldsymbol{p}_i) = -V_2(\boldsymbol{\nu}_e, \boldsymbol{z}_2) + \rho_c h_c(\boldsymbol{p}, \boldsymbol{p}_i) + \rho_c \iota_c(\|\epsilon\|_\infty)$.

Theorem 1: Consider the MASS model in (4) and (5), the sets Ω_s and Ω_v in (49) and (50), safe velocity satisfying (44), and the kinetic tracking control law (24). If $\mu > \max\{\gamma_o, \gamma_j, \gamma_c\}$, then the input-to-state safety is guaranteed such that $(\boldsymbol{p}_0, \boldsymbol{\nu}_e(0)) \in \Omega_v \Rightarrow p(t) \in \Omega_s, \forall t \geq 0$, where $\rho_* = (\mu - \gamma_*)/(\sqrt{2} C_{h*}) > 0$ and $\iota_*(\|\epsilon\|_\infty) = l(\|\epsilon\|_\infty)/(\rho_* \gamma_*)$ with $l(\|\epsilon\|_\infty) = \max\{\|\boldsymbol{PB}\|/\sqrt{2\lambda_{\min}(\boldsymbol{P})}, 1/\sqrt{2}\}\|\epsilon\|_\infty$ and $*$ denoting o, j and c.

Proof: Since $(p_0(0), \nu_e(0)) \in \Omega_v$ and $\dot{\hbar}_*(\cdot)$ satisfies

$$\begin{aligned}
\dot{\hbar}_*(\cdot) =& -\dot{V}_2 + \rho_* \frac{\partial h_*(\cdot)}{\partial \boldsymbol{p}} \underline{\boldsymbol{R}}(\boldsymbol{\psi}) \boldsymbol{q} \\
\geq& \ \mu V_2 - l(\|\epsilon\|_\infty) + \rho_* \frac{\partial h_*(\cdot)}{\partial \boldsymbol{p}} \underline{\boldsymbol{R}}(\psi)(\boldsymbol{q}_s + \boldsymbol{q}_e) \\
\geq& \ \mu V_2 - l(\|\epsilon\|_\infty) - \rho_* \gamma_* h_*(\cdot) + \rho_* \frac{\partial h_*(\cdot)}{\partial \boldsymbol{p}} \underline{\boldsymbol{R}}(\psi) \boldsymbol{q}_e \\
\geq& \ (\mu - \gamma_*) V_2 - \gamma_* \hbar_*(\cdot) - \rho_* \left\| \frac{\partial h_*(\cdot)}{\partial \boldsymbol{p}} \right\| \|\underline{\boldsymbol{R}}(\psi) \boldsymbol{q}_e\| \\
\geq& -\gamma_* \hbar_*(\cdot),
\end{aligned} \tag{51}$$

with $\boldsymbol{q}_e = \boldsymbol{q} - \boldsymbol{q}_s$, it can be concluded that $p(t) \in \Omega_v, \forall t \geq 0$. As $V_2(\boldsymbol{\nu}_e, \boldsymbol{z}_2) \geq 0$, it renders $\hbar_*(\cdot) > 0 \Rightarrow \rho_* \bar{h}_*(\cdot) > 0$. As a result, $\bar{h}_*(\cdot) > 0, \forall t \geq 0$, i.e., $p(t) \in \Omega_s, \forall t \geq 0$. ■

IV. Simulation Results

This section presents the simulation results to show the efficacy of the proposed safety-critical constrained anti-disturbance control method for a marine vehicle. The whole simulation is composed of ship model, kinematics control, dynamics control part, neurodynamic optimization, ocean disturbance and disturbance estimation. Consider a large marine vessel with its overall length being 76.2 m [39]. The model parameters during low-speed operations are given as follows:

$$\begin{cases} \boldsymbol{M} = 10^9 \times \begin{bmatrix} 0.0068 & 0 & 0 \\ 0 & 0.0113 & -0.0340 \\ 0 & -0.0340 & 4.4524 \end{bmatrix}, \\ \boldsymbol{D} = 10^8 \times \begin{bmatrix} 0.0008 & 0 & 0 \\ 0 & 0.0025 & -0.0203 \\ 0 & -0.0067 & 3.8481 \end{bmatrix}. \end{cases}$$

The nominal inertia matrix $\boldsymbol{M}^*$ may be estimated through experiments. For an MASS, its nominal parameters may be available from the ship-builder's specifications. In the simulation, $\boldsymbol{M}^*$ is set as:

$$\boldsymbol{M}^* = 10^9 \times \begin{bmatrix} 0.0068 & 0 & 0 \\ 0 & 0.0113 & 0 \\ 0 & 0 & 4.4524 \end{bmatrix}.$$

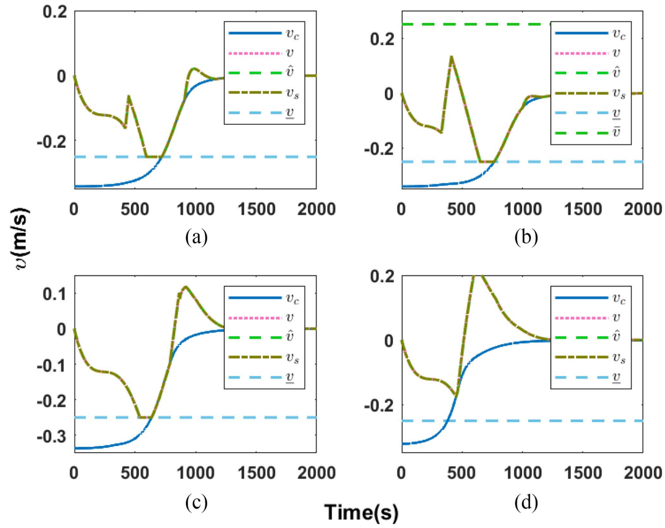

Fig. 10. Commanded signals, actual signals, estimated signal, and optimized signal of the sway velocity. (a) Case i, (b) Case ii, (c) Case iii, (d) Case iv.

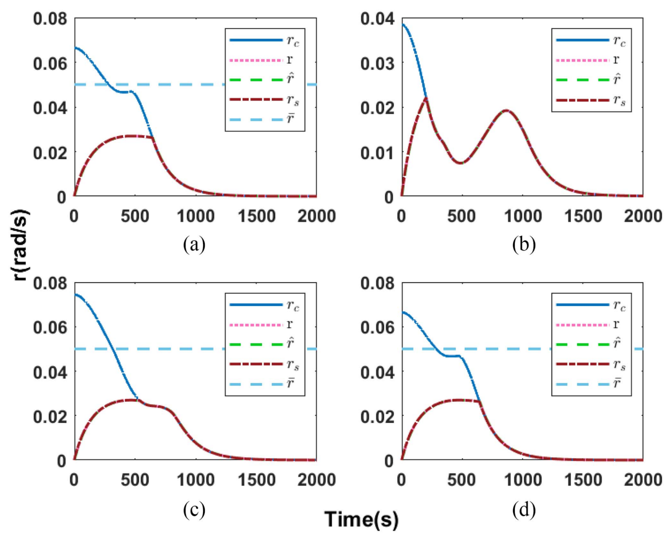

Fig. 11. Commanded signals, actual signals, estimated signal, and optimized signal of the yaw rate. (a) Case i, (b) Case ii, (c) Case iii, (d) Case iv.

The ocean disturbance is generated by a wind model. According to the reference [40], the wind forces and moments can be modelled as

$$\tau_{\text{wind}} = \frac{1}{2} \rho_a v_w^2 \begin{bmatrix} C_X(\gamma_w) A_{Fw} \\ C_Y(\gamma_w) A_{Lw} \\ C_N(\gamma_w) A_{Lw} L_{oa} \end{bmatrix} \tag{52}$$

where $v_w \in \Re$ is relative wind speed, $\gamma_w \in \Re$ is the angle of attack. A_{Lw} is the ship lateral projected area, A_{Fw} is the ship frontal projected area, ρ_a is the density of air at 20 degrees Celsius. v_w and γ_w are modelled as a first-order Gauss-Markov process:

$$\dot{v}_w + \mu v_w = w_u, \dot{\gamma}_w + \mu \gamma_w = w_\psi \tag{53}$$

where w_u and w_ψ are Gaussian white noise; $\mu \geq 0$ is a constant. The parameters for the wind model in (52) and (53) are listed in

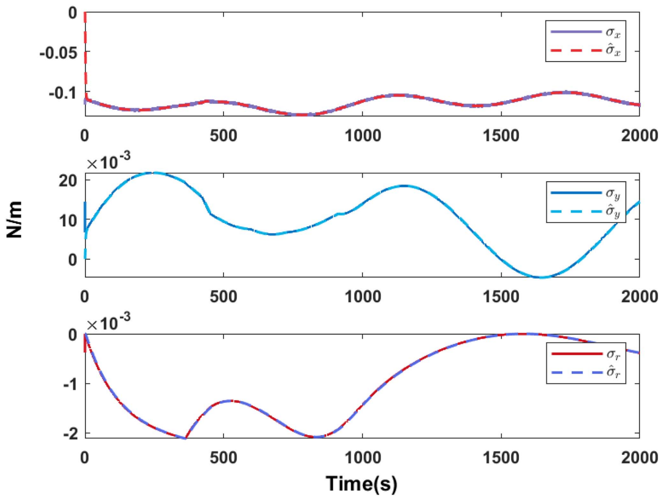

Fig. 12. Actual disturbance and estimated disturbance in Case i.

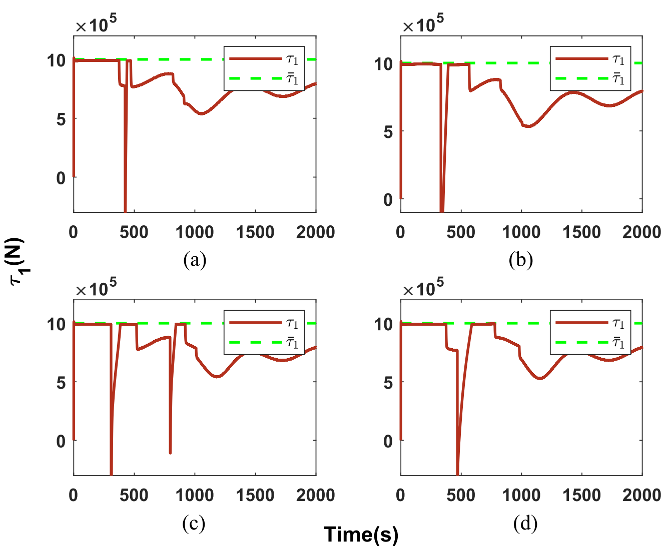

Fig. 13. The control force in the surge direction. (a) Case i, (b) Case ii, (c) Case iii, (d) Case iv.

Table I and the parameters C_X, C_Y, and C_N are selected as the same as [40].

A parameter selection guideline is provided as follows. k_1, k_2, and k_3 relate to the ship speed and they can be selected according to the maximal allowed velocity. The parameters Δ, Δ_1, Δ_2, and Δ_3 determine the transient performance of the automatic berthing. The larger values of Δ, Δ_1, Δ_2, and Δ_3 may result in a smoother transient and a longer convergence time. Hence, a trade-off should be made. The collision avoidance parameters γ_o, γ_j, and γ_c relate to the action range of control barrier functions. A small value of γ_o, γ_j, and γ_c leads to a large collision avoidance radius. A_{Fw}, A_{Lw}, and L_{oa} are selected according to the ship parameters. The parameter values are listed in Table I.

The Hong Kong-Macau Ferry Terminal is chosen as the berthing harbor and the reference frame origin is set to $[114^\circ 16' 90.78'' E, 22^\circ 29' 37.02'' N]$. Four cases from different initial positions are chosen to demonstrate automatic berthing.

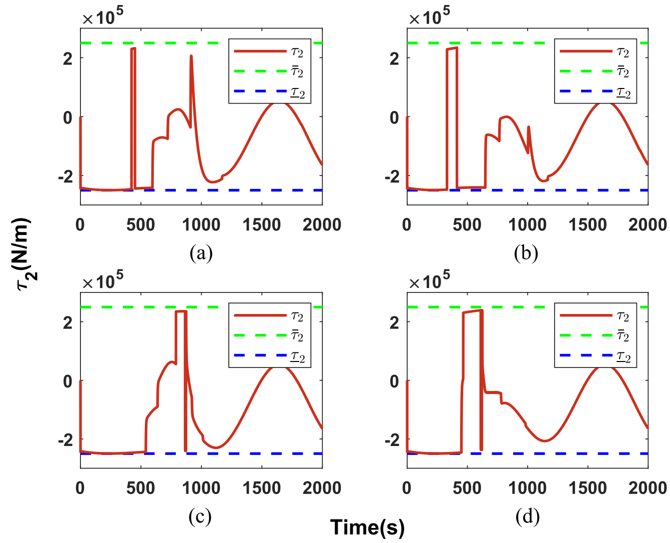

Fig. 14. The control force in the sway direction. (a) Case i, (b) Case ii, (c) Case iii, (d) Case iv.

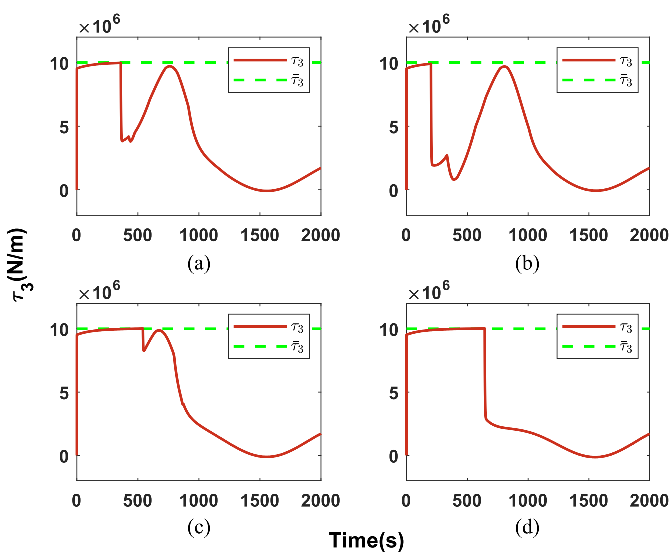

Fig. 15. The control force in the yaw direction. (a) Case i, (b) Case ii, (c) Case iii, (d) Case iv.

The initial positions of the marine vessel for the four cases are listed in Table II.

For the purpose of comparison, we select a baseline with the following automatic berthing control law without considering constraints as follows:

$$\begin{bmatrix} u_c \\ v_c \\ r_c \end{bmatrix} = \boldsymbol{R}^T(\psi_e) \begin{bmatrix} -\frac{k_1 s}{\sqrt{s^2+\Delta_1^2}} \\ -\frac{k_2 e}{\sqrt{e^2+\Delta_2^2}} \\ -\frac{k_3 \psi_e}{\sqrt{\psi_e^2+\Delta_3^2}} \end{bmatrix}. \tag{54}$$

In the following figures, subfigures (a), (b), (c), (d) show the simulation results for Case i, Case ii, Case iii, and Case iv. Fig. 5 depicts the automatic berthing trajectories of the MASS steered using the proposed safety-critical constrained anti-disturbance control method in the presence of different initial positions. It

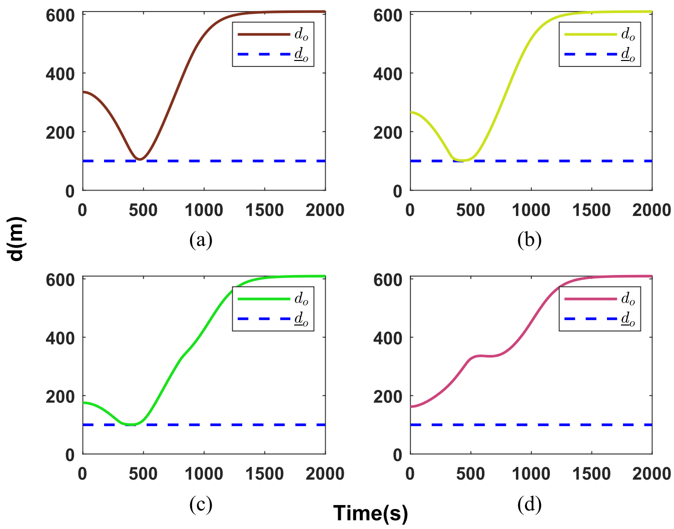

Fig. 16. The distance between the marine vessel and the obstacle. (a) Case i, (b) Case ii, (c) Case iii, (d) Case iv.

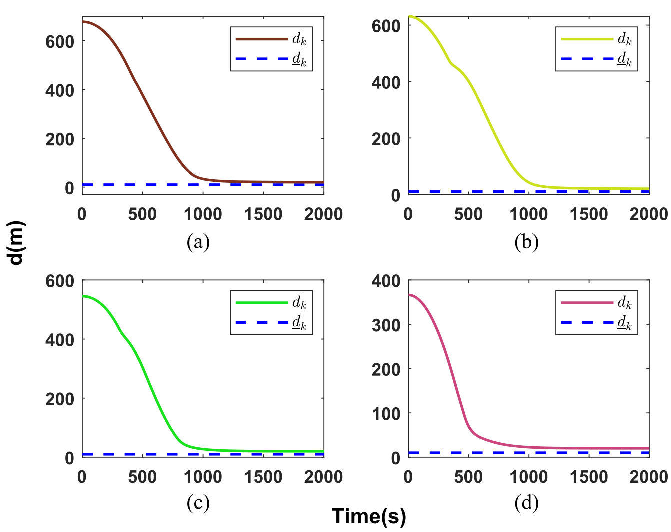

Fig. 17. The distance between the marine vessel and the shoreline. (a) Case i, (b) Case ii, (c) Case iii, (d) Case iv.

can be seen that successful berthing is achieved from four different initial positions in the presence of the ocean disturbance. Fig. 6 shows the berthing trajectories of the MASS by using the automatic berthing control law in (54), implying the inability of the control law in (54) for safety operations. The minimum distances relative to the obstacle and shoreline are presented in Table III. It can be seen that the collision-avoidance constraints are not violated by using the proposed safety-aware automatic berthing control law (46). Fig. 7 and Fig. 8 depict the position and heading tracking errors, respectively, which converge to a small neighborhood of zero. Figs. 9–11 depict the commanded surge speed, sway speed, and yaw rate u_c, v_c, r_c, the real surge speed, sway speed, and yaw rate u, v, r, the estimated surge speed, sway speed, and yaw rate $\hat{u}, \hat{v}, \hat{r}$, and safe surge speed, sway speed, and yaw rate u_s, v_s, r_s for the four cases. It is observed that the safe surge speed, sway speed, and yaw rate are within the constraints. Fig. 12 depicts the estimation performance of the ESO in the Case i simulation, showing that the ocean disturbance and model nonlinearities are accurately estimated by using the ESO. Figs. 13–15 depict the control force and yaw moment within bounds and constraints. Figs. 16–17 depict the distance between the MASS and the obstacle/shoreline, showing that minimal collision avoidance distance is ensured.

For comparison, Table III lists the obstacle-avoidance performance of the proposed control law (46) and the control law in (54), where $d_o = \sqrt{(x-x_o)^2 + (y-y_o)^2}$ is the distance between the geometric center of the ship and the stationary obstacle, $d_j = (\sqrt{(x-x_j)^2 + (y-y_j)^2}$ is the distance between geometric center of the ship and the kinetic obstacles, $d_k = (a_k x + b_k y + c_k)/\sqrt{(a_k)^2 + (b_k)^2}$ be the perpendicular distance between the geometric center of the ship and the shoreline. It shows that a collision takes place in Case iv, if control law (54) is used.

V. Conclusions

In this article, we present an optimization-based control method for MASS automatic berthing in a constrained water region with stationary and moving obstacles. The velocity constraints, input constraints, shoreline constraints, and collision avoidance requirements are incorporated into the automatic berthing control law design and command optimization. The proposed control system includes a kinematic guidance loop, an anti-disturbance kinetic control loop, and a command optimization loop. The proposed control method is able to enforce the satisfaction of the velocity constraints, input constraints, collision-avoidance constraints, in addition to maintaining stability high-performance against disturbances and other uncertainties. Simulation results with a large marine vessel substantiate the efficacy of the proposed safety-certified control method for automatic berthing subject to physical and environmental constraints.

The proposed automatic berthing control law is applicable to fully-actuated ships only. An extension to underactuated ships is more beneficial. In addition, the control parameters of the proposed automatic berthing control law are tuned via "trial and error" and it is desirable to use machine learning techniques to optimize control performance with respect to the parameters. Lastly, due to the unavailability of a large marine vessel, only simulation results are presented. In-field experiment verifications are more desirable and convincing. Further investigations may include the extension of the proposed automatic berthing control law to under-actuated vessels, performance-driven parameter optimization, and in-field experiments based on available marine vessels.

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